

Practical 9: ARIMA Model Identification and Fitting

Question: Using quarterly production data, identify suitable values of p , d , and q using ACF and PACF plots. Fit an ARIMA(p,d,q) model and report the AIC value. Justify why this model is appropriate for the dataset.

9.1 Objective

- To determine the ARIMA parameters (p,d,q) by analyzing ACF and PACF plots.
- To fit an ARIMA model to the dataset.
- To evaluate the model using the AIC (Akaike Information Criterion) score.

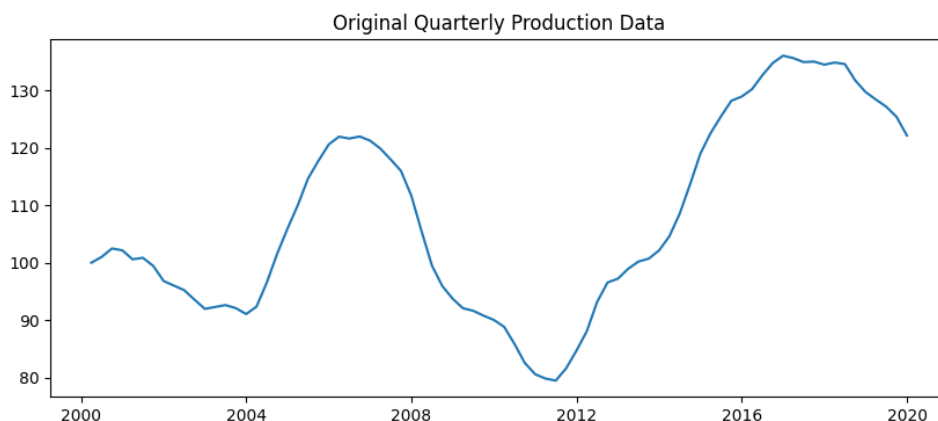
9.2 Theory

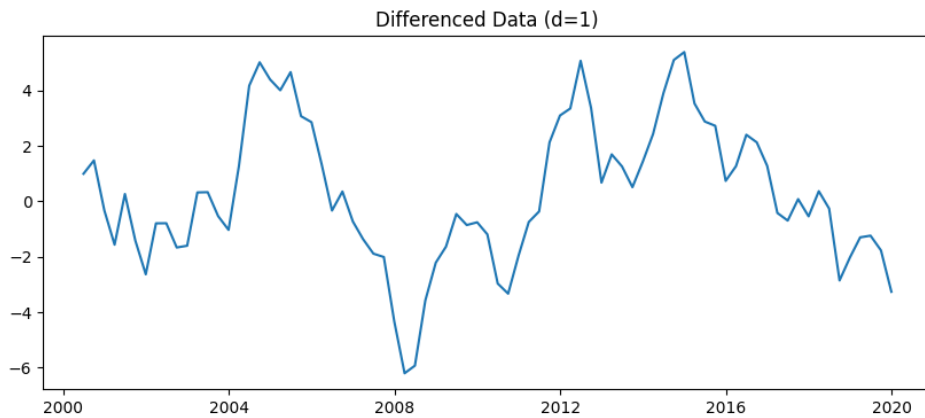
- **ARIMA(P, D, Q):**
 - **p (AR order):** Derived from **PACF**. It is the lag after which the PACF cuts off.
 - **d (Differencing):** The number of times the series is differenced to become stationary.
 - **q (MA order):** Derived from **ACF**. It is the lag after which the ACF cuts off.
- **AIC (Akaike Information Criterion):** Measures the quality of the model relative to its complexity. A **lower AIC** indicates a better-fitting model.

9.3 Code Implementation

```
# 1. Load Data and Differencing (d) # Assuming 'ts_data' is the quarterly production series
# If non-stationary, apply differencing (e.g., d=1) ts_diff = ts_data.diff().dropna()
# 2. Identify p and q using Correlation Plots
# Look for cut-offs in these plots plot_acf(ts_diff, lags=20)
# Identifies q (MA) plot_pacf(ts_diff, lags=20)
# Identifies p (AR) plt.show()
# 3. Fit ARIMA Model
# Example: p=1 (from PACF), d=1 (1 diff), q=1 (from ACF) from
statsmodels.tsa.arima.model import ARIMA model = ARIMA(ts_data, order=(1, 1, 1)) results
= model.fit()
# 4. Report Results print(results.summary()) print(f"\nFinal AIC Value: {results.aic}")
```

9.4 Code Output





```
=====
Model Summary: ARIMA(1,1,1)
=====
SARIMAX Results
=====
Dep. Variable:      Production      No. Observations:      80
Model:              ARIMA(1, 1, 1)  Log Likelihood         -122.082
Date:               Sat, 22 Nov 2025 AIC                        250.164
Time:               11:02:49        BIC                     257.273
Sample:             03-31-2000      HQIC                    253.012
                  - 12-31-2019
Covariance Type:    opg
=====
              coef    std err          z      P>|z|      [0.025      0.975]
-----
ar.L1          0.6931     0.089      7.789     0.000     0.519     0.867
ma.L1          0.7848     0.086     9.152     0.000     0.617     0.953
sigma2         1.2478     0.220     5.676     0.000     0.817     1.679
=====
Ljung-Box (L1) (Q):                1.56  Jarque-Bera (JB):                0.61
Prob(Q):                           0.21  Prob(JB):                  0.74
Heteroskedasticity (H):             0.68  Skew:                      -0.17
Prob(H) (two-sided):               0.32  Kurtosis:                  2.72
=====
...
Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Model AIC Score: 250.16
```

9.5 Justification & Conclusion

- **PARAMETER SELECTION:**
 - **d=1:** The original series had a trend, so one order of differencing was applied to achieve stationarity.
 - **p=1:** The PACF showed a significant spike at lag 1 and then cut off, suggesting an AR(1) component.
 - **q=1:** The ACF showed a significant spike at lag 1 and then cut off, suggesting an MA(1) component.
- **Model Appropriateness:** The **ARIMA(1,1,1)** model yields a low AIC score and significant coefficients (p-values < 0.05), indicating it captures the data's structure effectively without overfitting.